Pham Nhat Tien

SOFTWARE ENGINEER

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SUMMARY

Software Engineer with 3.5 years at FIDT, from client-product frontend architecture to core R&D platforms. Adapts fast and works independently with agentic AI workflows to ship innovations with business value.

TECH STACK

Frontend	Vue 3, Nuxt, TypeScript, React, Vite, SSR & hydration, Module Federation, SCSS, Flutter
Backend	Node.js, Bun, Hono, Zod, MongoDB, PostgreSQL, Redis, Directus, OIDC/OAuth2
AI & Tooling	LLM agents, MCP, GitLab CI

WORK EXPERIENCE

R&D Developer, FIDT — 02/2025 – Present

Research and build the core systems that product teams depend on.

Magisk — visual page builder (Figma-like, low-code)

- **Real-time collaboration:** authored the ADR for multi-user editing and version history: shadow-state diffing (0.03 ms per edit), per-field last-writer-wins, write-ahead journal and kill switch; rejected CRDTs and page locking.
- **Publishing pipeline:** phased move from client- to server-side rendering behind a flag, verified by screenshot-parity tests, with per-route cache, versioned invalidation and sandboxed page logic; save-to-live cut to ~9 s.
- **AI agent:** turns user requests, from text prompts to Figma designs, into editable pages; made reliable with tool-call validation, repeated-call detection, a contrast gate and a screenshot evaluator loop.
- **Agent evaluation (TGB):** designed and built Trace → Goal → Bench, an evaluation framework for the agent: trajectory traces, explicit goals, automated benchmarks and code-based quality judges with hard gates that catch reward hacking. Cut tool-call turns per build by 61% (140 → 55).

FIDT IAM — identity & access management

- Built the company's OpenID Connect provider on node-oidc-provider with Azure AD SSO, using Directus as the identity broker for users and roles.
- Implemented client credentials, device flow, token revocation and RP-initiated logout, JWKS key rotation, and token invalidation when a user's roles change.

FML — financial modelling DSL & calculation engine

- Designed a domain-specific language for personal financial statements: grammar defined with Ohm-js, a parser producing an AST, and an interpreter supporting conditionals, case statements, built-in financial functions and attribute access.
- Built a multi-period engine that derives net assets, net income, cash flow and cash for each month and rolls results up to quarters and years with type-specific aggregation rules; covered by Jest unit tests.

FIDT MVVM — frontend architecture framework, co-designed with the CTO

- Evolved traditional MVVM into a Vue implementation: views only bind, ViewModels hold reactive state and business logic, and EventStores own watchers and lifecycle.
- Adopted across all FIDT systems; documented as conventions and agent skills, and taught through workshops, hands-on sessions and code review.

Resource Guru → Saul — authorization service & API gateway (v1)

- Built Resource Guru, the FIDT app's authorization service, and evolved it into Saul.
- Designed and implemented Saul end to end: one YAML definition per service generates types, validation, client SDKs and permissions, enforced at the gateway.

FIDT UI Components — shared component library & design tokens

- Shared Vue component library used across product teams, plus an automated pipeline from Figma tokens to versioned npm packages with a type-safe theme.

Software Developer — Product Delivery, FIDT 04/2023 – 01/2025

Frontend standards & mentoring

- Set frontend conventions and project structure; reviewed code and trained interns and new engineers.

Wealth Platform foundation · Sales Platform advisor

- Built the foundation of the Wealth Platform: the initial standard structure for both frontend and backend that the team built on. Advised the Sales Platform team on architecture.

FIDT Portal — micro frontends

- Built and maintained FIDT Portal V1/V2 micro frontends: migrated from iframes to Module Federation, with shared state, cross-app messaging and performance tuning.

PERSONAL PROJECTS

Studio — multi-agent AI development framework 2026

- **Multi-agent orchestration:** a planner agent (Claude) delegates to PM, UX, dev, QA and QC agents running in parallel on lower-cost models, from approved spec to reviewed, browser-tested slices.
- **Sandboxed execution:** each project runs in isolated Docker with filesystem and egress allow-lists; results verified by tests. Used on a production Magisk feature with six parallel agents.

One Pane — tmux client for Android 2026

- Built for a personal need, efficient tmux over SSH on a phone: designed and developed with AI a Flutter app that speaks tmux control mode, one pane at a time; preparing for Google Play.

SKILLS

System design	Distributed sync, caching, API gateways, OIDC & JWT, DSLs & code generation
AI engineering	Agent architecture, multi-agent, agent evaluation (TGB), guardrails, sandboxing, MCP

EDUCATION

Ho Chi Minh City University of Technology and Education (HCMUTE)
Bachelor of Engineering — Software Engineering · Languages: Vietnamese (native), English (working)